Unit 1 Pre-Test

Equations, Inequalities & Reasoning

Standards assessed: NC.M1.A-SSE.1 · A-CED.1 · A-CED.4 · A-REI.1 · A-REI.3 · N-RN.2

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.A-SSE.1 · NC.M1.A-REI.3 · NC.M1.N-RN.2 Read expressions, solve one- and two-step equations, and use simple exponent rules.

• *I can name parts of an expression, solve simple equations and inequalities, and use exponent rules.*

Answer each question. Use the picture to help you.

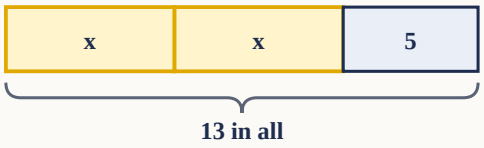
How to type your answers

x = 4 `4` negative `-7` power `x^5` inequality `x>2 or x<=-3`

1. *Solve*

$2x + 5 = 13$

not to scale



13 in all

x = _____

answer: x = 4

2. *Solve*

$\frac{x}{3} - 4 = 2$

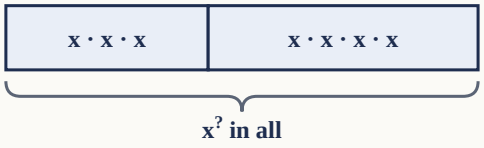


x = _____

answer: x = 18

3. *Simplify*

$x^3 \cdot x^4$



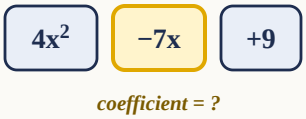
$x^?$ in all

Answer: _____

answer: x^7

4. *Parts of an expression*

In $4x^2 - 7x + 9$, what is the **coefficient** of x?



coefficient = ?

Answer: _____

answer: −7

5. *Solve the inequality*

$x + 4 > 10$

Graph your answer here



Answer:

answer: $x > 6$

6. *Simplify*

$$\frac{x^8}{x^2}$$

$$\frac{x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x}{x \cdot x} = x^{?}$$

Answer:

answer: x^6

SCORE 4.0 Target Content

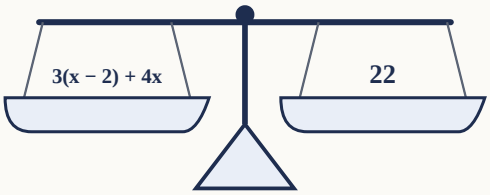
NC.M1.A-REI.1 · NC.M1.A-REI.3 · NC.M1.A-CED.1 · NC.M1.A-CED.4 · NC.M1.N-RN.2 Solve multi-step equations and inequalities, justify steps, and rearrange formulas.

• *I can solve multi-step equations and inequalities, explain each step, write equations from words, and solve a formula for one letter.*

Solve each problem. Show your work on paper.

7. *Solve*

$3(x - 2) + 4x = 22$



x =

answer: $x = 4$

8. *Solve the inequality*

$-3x + 4 \geq 13$

Graph your answer here

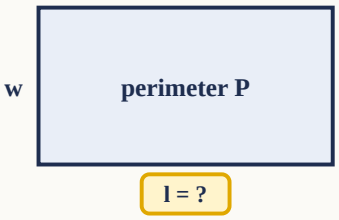


Answer:

answer: $x \leq -3$

9. *Literal equation*

The perimeter of a rectangle is $P = 2l + 2w$. Solve for l.



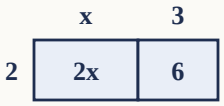
$l =$ _____

answer: $l = \frac{P - 2w}{2}$ (or $l = \frac{P}{2} - w$)

10. *Justify the step*

$2(x + 3) = 14 \rightarrow 2x + 6 = 14$

Which property makes this step true?



Answer: _____

answer: the distributive property

11. *Write and solve*

A gym charges $20 to join plus $15 each month. Kai paid $95 in all. How many months did he pay for?

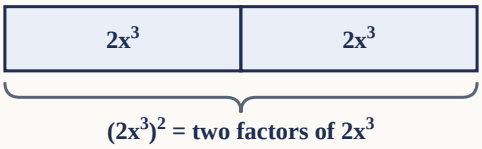


months = _____

answer: 5 months

12. *Simplify*

$(2x^3)^2$



Answer: _____

answer: $4x^6$

SCORE 5.0 Advanced Content

NC.M1.A-SSE.1 · NC.M1.A-REI.3 · NC.M1.A-CED.1 · NC.M1.A-CED.4 · NC.M1.N-RN.2 Solve harder equations, interpret models, and use negative exponents.

• *I can clear fractions, find special solutions, interpret parts of a model, and rewrite negative exponents as positive ones.*

Solve each problem. Some answers are words.

13. *Solve*

$$\frac{x}{2} + 3 = \frac{x}{3} + 5$$

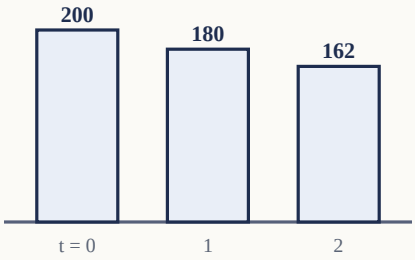


x = _____

answer: x = 12

14. *Interpret*

A model is $P = 200(0.9)^t$. By what **percent** does P go down each year?



percent = _____

answer: 10%

15. *Literal equation*

Solve y = mx + b for x.

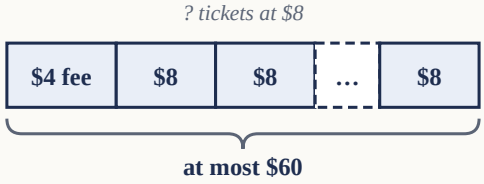


x = _____

answer: $x = \frac{y - b}{m}$

16. *Write and solve*

Concert tickets cost $8 each plus one $4 fee. Mia has $60. What is the **greatest** number of tickets she can buy?



tickets = _____

answer: 7 tickets

17. *Negative exponents*

Simplify $(2x^3)^{-2}$. Write your answer with **positive** exponents only.

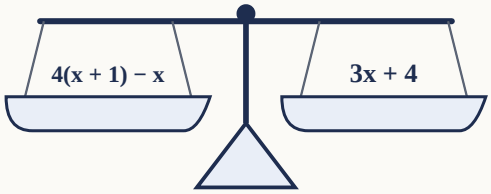
$$a^{-n} = \frac{1}{a^n}$$

Answer:

answer: $\frac{1}{4x^6}$

18. *How many solutions?*

$4(x + 1) - x = 3x + 4$



Answer:

answer: infinitely many solutions